Thanicha Ruangmas

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Applied economist with 7+ years of experience assessing the distributional impacts of policy by integrating geospatial data with census tract data, including the environmental consequences of energy subsidy reform. Combines econometric methods with large language models to process large unstructured data sources in data-scarce, developing-country settings.

PROFESSIONAL EXPERIENCE

Global Environment Facility Independent Evaluation Office (GEF IEO) Washington, D.C.
Data Scientist (ETC) 02/2025 – Present

- Apply large language models (LLMs), geospatial data, and econometric analysis to inform findings and recommendations for evaluations of policy coherence, urban sustainability, and the Eighth Comprehensive Evaluation of the GEF (OPS8).
- Link LLM-based text classifications of project documents such as policy coherence intent with project outcome ratings, using semiparametric matching to estimate outcome differences across project groups.
- Assess whether projects in Uzbekistan and coastal West Africa are located where environmental needs are greatest, using LLMs to extract project locations and integrating remote sensing data (Hansen Global Forest Change, Terra MODIS water quality) into a spatial relevance index.
- Maintain reproducible workflows in Python and R (Jupyter, Quarto, Git).
- Deliver technical products for multiple concurrent evaluations under tight deadlines, including upcoming evaluations of gender policy, Caribbean Small Island Developing States, and West Africa country clusters.

University of Maryland College Park, MD
Assistant Clinical Professor in Sustainability Analytics 01/2019 – 01/2025

- Designed and taught a 3-semester research methods curriculum and led an undergraduate research lab to assess the impacts of environmental policies.
- Co-authored a peer-reviewed article on fostering research skills and presented the approach at three conferences.
- Served as Principal Investigator for a USDA-APHIS study leveraging the Google Places API and natural language processing to identify 1,566 poultry and egg farms. Co-authored a technical report and built an R Shiny dashboard mapping farms. Presented findings to over 100 scientists.

Environmental Protection Agency (EPA)
Peer Reviewer 01/2022 – 12/2022

- Provided expert feedback on EPA's Power Plant Environmental Justice Screening Methodology, which estimates each U.S. power plant's pollution burden on vulnerable demographic groups based on its dispersion patterns.
- Wrote a technical user guide on leveraging the National Oceanic and Atmospheric Administration's (NOAA's) pollution dispersion model and its concentration trajectories to identify vulnerable demographic groups.

University of Wisconsin–Madison Madison, WI
Graduate Research Assistant 01/2015 – 12/2016

- Co-authored a peer-reviewed article assessing the distributional impacts of California's emissions trading program by combining results from a pollution dispersion model and U.S. Decennial Census data. Estimated heterogeneous treatment effects with Conley spatial HAC standard errors to account for spatial correlation.
- Wrote a job market paper that assessed the air pollution consequences of natural gas subsidy reform in Thailand and presented it at five international conferences.

International Institute for Applied Systems Analysis (IIASA) Laxenburg, Austria
Young Scientists Summer Program Fellow 05/2014 – 08/2014

- Conducted input-output analysis in MATLAB to determine whether emissions reductions in the EU were driven by adopting pollution abatement technologies or by relocating production to other regions.

PwC Thailand Bangkok, Thailand
Associate, Financial Information Technology Consulting Services 01/2011 – 05/2012

- Conducted surveys for the Thai government and assessed the financial reporting data pipeline for one of Thailand's largest banks.

EDUCATION

Ph.D. in Agricultural and Applied Economics, University of Wisconsin–Madison 2018
B.A. in Economics, Chulalongkorn University, Bangkok, Thailand 2011

FELLOWSHIPS AND GRANTS

USDA-APHIS Summer Grant ($16k) 2021
USDA-APHIS Summer Grant ($15k) 2020
Global Health Institute Research Award, UW–Madison ($4k) 2016
Economy and Environment Program for Southeast Asia Grant ($10k) 2015
Young Scientists Summer Program Fellow, IIASA 2014
Undergraduate Fellowship, Chulalongkorn University 2007–2011

AWARDS AND CERTIFICATIONS

Faculty Mentor Award, Philip Merrill Presidential Scholars Program, University of Maryland 2022
LangChain for LLM Application Development 2026
Systematically Improving RAG Applications 2025

PEER-REVIEWED PUBLICATIONS

- **Ruangmas, T.** & Olson, L. J. (2025). FIRE Sustainability Analytics: An Innovative Approach to Engaging Undergraduate Students in Economics Research. *Applied Economics Teaching Resources, 6*. doi:10.71162/aetr.782095

- Grainger, C. & **Ruangmas, T.** (2018). Who Wins from Emissions Trading? Evidence from California. *Environmental and Resource Economics, 71*(3), 703–727. doi:10.1007/s10640-017-0180-1

TECHNICAL REPORTS

- **Ruangmas, T.** (2022). *How to Run HYSPLIT and Calculate Affected Population with R.* [Link]
- Chapman, J., Kothari, S., Lazarus-Hall, A., Olson, L. J., Relan, M. & **Ruangmas, T.** (2020). *Using Online Data to Identify Farms Selling Poultry and Eggs in Delaware, Maryland, and Virginia.* FIRE–USDA APHIS Report. [Link]
- **Ruangmas, T.** (2015). *Decomposition of SO_2 and NO_x Emissions in the EU: Leakage or Technology Adoption?* IIASA YSSP Proceedings.

CONFERENCE PRESENTATIONS

- **Ruangmas, T.** (2024, June 25). *Fostering Data Literacy through Paper Replication & Reproducible Workflows.* ConnectUR Conference 2024, College Park, MD. [Slides]
- **Ruangmas, T.** (2024, May 31). *Fostering Data Literacy through Paper Replication & Reproducible Workflows.* CTREE Conference 2024, Atlanta, GA.
- **Ruangmas, T.** (2023, January 7). *Empirical Training for a Successful Course-Based Undergraduate Research Experience* [Poster]. ASSA 2023 Annual Meeting, New Orleans, LA.
- Chapman, J., Kothari, S., Lazarus-Hall, A., Olson, L. J., Relan, M. & **Ruangmas, T.** (2020, August 28). *Using Online Data to Identify Farms Selling Poultry and Eggs in Delaware, Maryland, and Virginia.* USDA-APHIS Virtual Science Webinar.
- **Ruangmas, T.** (2016–2017). *The Consequences of Natural Gas Subsidy Reform: The Case of Air Pollution in Thailand.* Presented at EAAERE (Singapore), IAEE (Singapore), EEPSEA (Bangkok), ICTH (San José, CA), and AERUS (Urbana-Champaign, IL).

TEACHING

University of Maryland, College Park

- FIRE120: FIRE Semester 1 2019–2024
- FIRE198: FIRE Semester 2 – Sustainability Analytics [Syllabus] 2019–2024
- FIRE199: Summer Internship – Sustainability Analytics [Syllabus] 2022–2024
- FIRE298: FIRE Semester 3 – Sustainability Analytics [Syllabus] 2020–2024

University of Wisconsin–Madison

- AAE244: Environment and the Global Economy (Teaching Assistant) 2015
- AAE343: Environmental Economics (Teaching Assistant) 2012–2014

SKILLS

Data Science & AI:	Python (LangChain, Pydantic, LanceDB, AsyncOpenAI)
Statistical Programming:	R (fixest, Synth), Stata
Geospatial Programming:	R (terra, ggmap), ArcGIS
Other:	Git/GitHub, Google Cloud Platform, Shiny, LaTeX, MATLAB
Languages:	English (fluent), Thai (native)
Status:	Thai national with U.S. Permanent Residency